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Neural Correlates of Belief Change in Political and Non-Political Domains Among Left-Wing Individuals Confronted with Counterarguments

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Abstract

This preregistered study investigates the neural mechanisms underlying resistance to belief change among individuals with strong left-wing orientations, examining both political and non-political domains. Building on previous work[1], we aimed to assess whether the neural signatures of belief resistance are culturally universal by replicating the study in a non-American context. Forty-three participants who held firmly established left-wing views were presented with counterarguments targeting their political and non-political beliefs while undergoing functional magnetic resonance imaging (fMRI). Consistent with the original study, participants exhibited robust resistance to changing their political beliefs, although demonstrating greater openness to revising non-political ones. Challenges to political beliefs elicited heightened activation within the Default Mode Network (DMN), especially in regions associated with self-referential processing and introspection—highlighting the identity-protective function of such beliefs. Moreover, greater resistance to belief change was linked to increased activation in the dorsomedial prefrontal cortex (dmPFC) and reduced activity in the orbitofrontal cortex (OFC), a region implicated in belief updating. In contrast to the original findings, we observed no correlation between belief change and (lower) activation in emotion-related regions such as the insula and amygdala. Taken together, these results suggest that the process of belief change—across both political and non-political domains—is deeply rooted in neural systems responsible for maintaining self-identity.

Keywords: left-wing beliefs, political and non-political beliefs change, neural correlates, conceptual replication

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Introduction

At the core of human reasoning lies an apparent paradox. Although rational thought is often conceptualized as a mechanism for forming accurate beliefs and updating them in light of new evidence, decades of psychological and behavioral research reveal a very different reality: people seldom revise their beliefs when confronted with disconfirming information [2,3,4,5]. Rather than functioning as impartial evaluators of evidence, individuals tend to engage in a form of *motivated cognition*—assimilating new information in ways that safeguard pre-existing worldviews, moral commitments, and social identities[6,7,8,9].

This tendency is especially pronounced when the beliefs in question are politically or morally charged [10,11]. Politicized issues such as abortion, LGBTQ+ rights, or immigration are rarely evaluated on the basis of purely evidentiary grounds; instead, they are interpreted through the prism of ideological commitments and identity-based affiliations [12,13]. In these contexts, well-documented cognitive biases—including ~~confirmation bias,~~ myside bias, and identity-protective reasoning—come into play, enabling individuals to selectively attend to, reinterpret, or outright reject information that poses a threat to their established convictions [14,15]. Such resistance is not merely intellectual; it is often deeply emotional and visceral, reflecting the fact that, for many people, political beliefs function not just as opinions but as integral components of self-definition [8,16,17].

This insight has been corroborated by emerging research in social and cognitive neuroscience [18,19,20,21]. A seminal study by Kaplan et al. [1] employed functional magnetic resonance imaging (fMRI) to explore the neural mechanisms underlying resistance to belief change. Their findings showed that challenges to deeply held political beliefs elicited heightened activation in brain regions associated with self-referential processing, the Default Mode Network

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(DMN). Strikingly, this neural activity was markedly reduced when participants encountered counterarguments directed at non-political beliefs, indicating that political beliefs are uniquely insulated by psychological defense mechanisms. Kaplan and colleagues further observed that greater resistance was associated with increased activity in the dorsomedial prefrontal cortex (dmPFC) and reduced activity in the orbitofrontal cortex (OFC), a region involved in belief updating. Conversely, participants more amenable to changing their views displayed lower activation in emotion-related areas such as the insula and amygdala. Collectively, these findings suggest that resistance to belief change may not stem from ignorance or deficient reasoning ability, but rather from the brain's intrinsic efforts to protect the self.

Although these findings were groundbreaking, their scope was limited. Conducted in the United States—a two-party system with relatively stable ideological boundaries—Kaplan et al.'s study raises an important question: do these neural responses generalize to other sociopolitical contexts where ideological divisions are shaped by different historical trajectories, cultural values, and institutional dynamics? Furthermore, although neuroimaging offers powerful insights, it also imposes constraints on ecological validity and scalability. To more fully understand belief resistance in a more diverse set of contexts, it is essential to broaden both the methodological toolkit and the populations under study.

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The present preregistered study addresses this gap by replicating and extending Kaplan et al.'s paradigm in the distinct sociopolitical setting of Poland—a society marked by a hybrid of democratic consolidation and democratic backsliding, shaped by its post-communist transformation, enduring historical cleavages, and escalating political polarization [22]. In this context, we investigate whether the same neural patterns of belief resistance emerge across different issue domains, specifically by comparing politically charged statements with non-

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political assertions. Political beliefs are typically normative, reflecting what individuals think *should* happen in society—for example, "Abortion should be legal" or "The government should promote LGBTQ+ rights." Non-political beliefs, by contrast, are empirical statements about observable facts, such as "Taking a multivitamin improves one's health" or "Second-hand smoke increases the risk of lung disease." This distinction matters because political beliefs are closely tied to identity and values, making them potentially more resistant to change, whereas non-political beliefs can often be updated based on evidence. In the experiment, participants were presented with counterarguments for both types of beliefs. For political beliefs, the additional information included normative arguments drawn from socio-political discourse in Poland, designed to challenge participants' positions while respecting their identity. For non-political beliefs, participants received empirical evidence directly contradicting their prior stance. Based on previous findings [1], we expected that this information would prompt greater revision of non-political beliefs, while political beliefs would remain more stable, reflecting the influence of identity and values on belief resistance, including effects observed at the neural level.

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We focused on individuals with strongly left-wing beliefs for both theoretical and methodological reasons. From a theoretical perspective, belief updating and resistance are most clearly observable when beliefs are firmly held and closely tied to identity. Individuals with strong ideological commitments are therefore the most informative population for testing hypotheses about the neurocognitive mechanisms underlying belief resistance. Studying less polarized individuals would increase variability in baseline conviction and potentially obscure the processes of interest. From a methodological standpoint, concentrating on a clearly defined ideological group allowed us to reduce heterogeneity related to political engagement, prior knowledge, and motivational investment, thereby increasing internal validity. This focus also enabled a close conceptual replication of Kaplan et al., who similarly examined participants with strongly left-

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wing beliefs, facilitating direct comparison across cultural contexts. Our study therefore serves both as a conceptual replication and a context-sensitive test of whether resistance to belief change is a universal psychological mechanism, or one that varies systematically across political cultures and information environments.

This investigation is particularly timely. As political polarization intensifies in many parts of the world, and as public discourse becomes increasingly fragmented by misinformation and ideological echo chambers, understanding the psychological and neural roots of belief rigidity is not merely an academic pursuit but a societal imperative. By examining how people process counterevidence in diverse political settings, this research contributes to a more global and nuanced account of belief persistence and lays the groundwork for designing more effective interventions aimed at fostering open-mindedness and strengthening democratic dialogue.

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Results

Behavioral results: Belief change

A repeated-measures MANOVA was conducted to examine the effects of time (before vs. after reading challenges) and statement type (political vs. non-political) on participants' belief ratings. The analysis revealed no significant main effect of statement type, Pillai's Trace = .004, $F(1, 42) = 0.165$, $p = .687$, partial $\eta^2 = .004$, indicating that—when time was not taken into account—belief strength did not significantly differ between political and non-political statements. However, there was a significant main effect of time, Pillai's Trace = .318, $F(1, 42) = 19.565$, $p < .001$, partial $\eta^2 = .318$, indicating that, regardless of statement type, participants' belief ratings changed significantly after reading the challenges, with an overall decrease in strength. Crucially, a significant interaction between time and statement type emerged, Pillai's Trace = .393, F(1,42)

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= 27.181, $p < .001$, partial $\eta^2 = .393$, showing that the magnitude of belief change over time depended on whether the statements were political or non-political.

Descriptive statistics clarify the pattern and give support for this interpretation. For non-political statements, mean belief ratings decreased from 6.13 (SD = 0.39) before the challenge to 5.37 (SD = 0.87) afterward, corresponding to a large effect size (Cohen's d = 1.13). For political statements, the decrease was negligible—from 5.72 (SD = 0.71) to 5.68 (SD = 0.66), corresponding to a negligible effect size (Cohen's d = 0.06). Thus, belief change was considerably more pronounced for non-political statements, highlighting greater resistance to belief updating in the political domain. This pattern was confirmed by pairwise comparisons using Bonferroni correction. The reduction in belief strength for non-political statements was statistically significant (mean difference = 0.756, SE = 0.136, $p < .001$, 95% CI [0.480, 1.031]), whereas the change for political statements was not significant (mean difference = 0.041, SE = 0.084, $p = .629$, 95% CI [–0.128, 0.210]). Taken together, these findings suggest that although belief strength generally decreases after exposure to counterarguments, this effect is driven almost entirely by changes in response to non-political statements. Political beliefs, by contrast, appear considerably more resistant to change. These results are illustrated in Figure 2.

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To assess the stability of the observed effects, a follow-up measurement was conducted approximately one week after the initial study. A repeated-measures MANOVA comparing belief strength at baseline (before the challenge) and at follow-up revealed a pattern consistent with the immediate post-test findings. There was no significant main effect of statement type, Pillai's Trace = .005, $F(1, 31) = 0.149$, $p = .702$, partial $\eta^2 = .005$, indicating that—when averaged across time—belief strength did not differ significantly between political and non-political statements. The main effect of time approached significance, Pillai's Trace = .116, $F(1, 31) = 4.081$, $p = .052$, partial η^2

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= .116, suggesting a general trend toward belief change over time. More importantly, a significant interaction between time and statement type was observed, Pillai's Trace = .223, $F(1, 31) = 8.893$, $p = .006$, partial $\eta^2 = .223$, indicating that the extent of belief change from baseline to follow-up depended on whether the statements were political or non-political. Descriptive statistics confirmed this pattern: belief strength for non-political statements declined from M = 6.07 (SD = 0.40) at baseline to M = 5.65 (SD = 0.76) at follow-up showing a medium-to-large effect (Cohen's $d \approx 0.69$), whereas political statements showed no meaningful change (Cohen's $d \approx 0.06$), increasing only slightly from M = 5.79 (SD = 0.70) to M = 5.83 (SD = 0.70).

Bonferroni-corrected pairwise comparisons further supported this result: the pre–follow-up difference between political and non-political statements was statistically significant (mean difference = 0.285, p = .043). No significant difference was found between the two types of statements at the follow-up measurement itself (p = .345), suggesting that the divergence arose from the change relative to baseline rather than from differences at a single time point. These results indicate that the belief change observed immediately after exposure to counterarguments was not transient. The greater reduction in belief strength for non-political statements, compared to political ones, persisted even after approximately one week, underscoring that resistance to updating political beliefs is not only robust but somewhat enduring.

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Behavioral results: Response Time

To examine the cognitive processing underlying belief revision, we also analyzed the time that participants took to rate their belief strength following the presentation of challenges. A paired-samples t-test revealed that participants responded significantly faster when evaluating political statements than when evaluating non-political statements after reading the challenges. Specifically, the average response time for political statements was M = 5.16 seconds (SD = 1.19),

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compared to M = 5.70 seconds (SD = 0.96) for non-political statements. This difference was statistically significant, t(42) = –3.53, p = .001, with a mean difference of –0.54 seconds (95% CI: [–0.85, –0.23]). These results suggest that participants required less time to evaluate political statements, potentially reflecting greater cognitive rigidity, reliance on heuristic processing, or reduced deliberation when engaging with political content—even in the face of counterevidence. By contrast, the longer response times for non-political statements may indicate more effortful cognitive engagement or greater openness to reconsideration in those cases.

Brain Imaging Results

In the whole-brain analysis, we compared neural activity during challenges to political versus non-political beliefs—our primary contrast of interest (Figure 3, Table 1). Processing challenges to political beliefs elicited relatively greater activation in regions of the Default Mode Network (DMN), including the posterior cingulate cortex, medial prefrontal cortex, and right supramarginal gyrus. In contrast, reading challenges to non-political statements was associated with increased activation in the attention network, including bilateral dorsolateral prefrontal cortices, orbitofrontal cortices, the inferior temporal gyrus, and—medially—the presupplementary motor area, relative to political challenges.

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We next examined whether item-level differences in belief change correlated with brain activity during the presentation of challenges to those items (Figure 3D depicts the degree of belief change across different statements). Three brain regions displayed activity that significantly correlated with belief persistence across items (Figure 4 A, B,C). Signal levels in the right inferior temporal gyrus (MNI +60,-47,-11) and left temporal pole (MNI -34, -2, -35) correlated positively with belief change scores, whereas activity in the dorsomedial prefrontal cortex (dmPFC; MNI -15, +47, +35) correlated negatively. It is important to note that the scatter plots in Figure 4 are provided solely for visualization purposes and do not constitute independent statistical analyses.

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In addition to these item-wise correlations, we also conducted subject-wise analyses examining the relationship between individual belief-change scores for non-political statements, and neural activity during the reading of challenges, measured within *a priori* defined regions of interest (ROIs), specifically, three subregions of the insular cortex, and the amygdala complex. A Shapiro-Wilk test showed that the signal change data from the amygdala, as well as the average belief-change scores for non-political statements, did not deviate significantly from a normal distribution. However, it also revealed that the distributions of beta values for all three insular regions differed significantly from normality.

Frequentist analyses revealed no significant correlations between belief-change scores and neural activity in any of examined ROIs (amygdala: $r = 0.12$, $p = 0.45$; dorsal anterior insula: $\rho = -0.13$, $p = 0.39$; posterior insula: $\rho = -0.02$, $p = 0.88$; ventral anterior insula: $\rho = -0.14$, $p = 0.38$). Illustrations of the effects are presented in Figures S1-S4 in the Supplementary Materials. Bayesian analysis further indicated, for all ROIs, moderate evidence for the absence of correlation between activity from these regions and subject-wise belief change (amygdala: $BF_{01} = 4.04$; dorsal anterior insula: $BF_{01} = 4.56$; posterior insula: $BF_{01} = 5.32$; ventral anterior insula: $BF_{01} = 4.17$).

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Discussion

Our study replicates and extends previous findings on belief resistance [1,18,19,20,21], demonstrating that challenges to political beliefs evoke distinct neural and behavioral responses compared to non-political beliefs—even within a markedly different cultural and political context. Participants with strong left-wing views showed greater resistance to belief change when confronted with political counterarguments than with non-political ones. This resistance manifested on both behavioral and neural levels. Behaviorally, it was evidenced by the smaller changes in belief ratings following political challenges. Neurally, it was marked by heightened

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activation within the Default Mode Network (DMN), particularly in regions associated with self-referential processing such as the medial prefrontal cortex and posterior cingulate cortex. These findings align with a growing body of evidence suggesting that political beliefs are processed not merely as objective, factual claims, but as integral components of the self-concept [23,24]. The observed activation in the dorsomedial prefrontal cortex (dmPFC)—a region linked to identity-protective cognition—mirrors the findings of Kaplan et al., underscoring the robustness of this neural mechanism across cultural contexts. By contrast, belief change in response to non-political challenges was associated with increased activation in brain regions supporting executive function and attentional control, including the orbitofrontal cortex (OFC) and dorsolateral prefrontal cortex (dlPFC). This pattern suggests greater engagement in cognitive reappraisal and belief-updating processes when the threat to one's identity is minimal.

By replicating these findings in Poland—a post-communist democracy with a political landscape and media environment markedly different from that of the United States—our study offers evidence for the cultural robustness of the neural mechanisms underlying belief resistance. Despite the distinct political history of Poland and the explicitly left-wing identification of our sample, we observed the same patterns of neural defensiveness when participants' core political beliefs were challenged. This convergence suggests that belief resistance is not solely a function of U.S.-style partisanship, but may represent a cross-cultural phenomenon grounded in the neurocognitive architecture of self-protection.

Although prior research has linked emotional reactivity—indexed by amygdala and insula activity—to resistance to belief change, our results showed no significant correlation between activity in these emotion-related ROIs and belief change in response to non-political challenges. Bayesian analyses further provided moderate evidence in favor of the null hypothesis, reinforcing

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the absence of such a relationship. Taken together, these findings point toward a predominantly cognitive, rather than affective, mechanism in this context. This finding both enriches and brings nuance to existing models by suggesting that self-relevance, rather than emotional arousal per se, may be the primary driver of belief rigidity, at least among individuals with left-leaning political orientations.

The shorter response times observed for political statements—despite their complexity and personal relevance—are quite indicative of a form of cognitive rigidity and/or reliance on heuristic processing. When evaluating challenges to their political views, participants appeared to engage in less deliberative processing, potentially due to prior commitments, motivational filtering of information, or a desire to resolve cognitive dissonance rapidly. Conversely, the longer deliberation observed for non-political statements likely reflects deeper cognitive engagement with, and more thorough consideration of, the presented evidence.

This pattern is notable because it contrasts with earlier findings showing longer reading times for counter-attitudinal political arguments, which have been interpreted as reflecting effortful scrutiny aimed at identifying flaws in opposing viewpoints[25]. One possible explanation is that, under conditions of high identity relevance and strong prior commitment, participants may disengage from analytic evaluation more quickly, defaulting instead to rapid dismissal heuristics rather than extended counterarguing. Alternatively, task demands or time pressure may have shifted processing from elaborative scrutiny toward faster, affect-driven judgment, suggesting that belief defense can manifest either as prolonged engagement or as accelerated rejection, depending on contextual and motivational factors Also, it is important to note that the political statements used in this study were normative “should” propositions rather than empirical claims, whereas the non-political statements were factual in nature. This difference could, in principle, confound

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observed differences in belief updating between political (non-empirical) and non-political (empirical) beliefs. As in the study by Kaplan et al., however, this distinction is theoretically meaningful rather than incidental. Political beliefs are closely tied to personal values and social identity, which makes them more resistant to change, whereas non-political beliefs are often more amenable to revision based on evidence. This interpretation is further consistent with findings by Howlett and Paulus [26], who distinguished between testable and non-testable beliefs and showed that evaluating these two types of propositions engages partially overlapping but also distinct neural systems. In their study, the posterior parietal cortex was more active when participants evaluated testable beliefs—corresponding to non-political beliefs in our paradigm—than non-testable beliefs, which map onto our political condition. Importantly, this pattern suggests that the non-political condition in both Kaplan's study and ours may, similar to Howlett and Paulus's testable condition, elicit deeper or qualitatively different forms of cognitive processing than their respective counterparts.

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Our findings carry several implications for theories of motivated reasoning, political cognition, and neural self-protection. First, they reinforce the idea that belief resistance is domain-specific, with political beliefs receiving more neural protection than other categories of beliefs. Second, they highlight the central role of self-referential brain networks, particularly the DMN, in shaping responses to counterevidence. Third, these findings suggest that belief updating may require more than mere exposure to countervailing information. The observed patterns point to the central role of identity-relevant commitments and self-protective processing in shaping responses to belief challenge. From a practical standpoint, the results caution against relying solely on rational, evidence-based appeals when engaging individuals on politicized issues. Instead, interventions aimed at promoting belief change may need to explicitly account for identity concerns and

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motivational factors that structure how information is filtered, evaluated, or dismissed, even in the absence of strong affective responses.

Several limitations should be acknowledged. First, our sample consisted exclusively of individuals with strongly held left-wing views, which necessarily limits the generalizability of the findings across the full ideological spectrum. We acknowledge that including right-wing or less ideologically polarized participants would allow for a more comprehensive examination of political belief updating. However, the present study constitutes an initial investigation in a new cultural context, and focusing on a relatively homogeneous ideological group enabled us to control for baseline differences in political engagement, prior knowledge, and motivational factors that could otherwise confound the results. Importantly, this design choice also allowed us to conceptually replicate and extend the work of Kaplan et al., who similarly focused on participants with strongly held left-wing beliefs. Maintaining this parallel facilitates more direct comparisons of both neural and behavioral patterns of belief resistance across studies. Moreover, examining individuals with firmly established political beliefs increases the likelihood of observing robust resistance to counterevidence, which is essential for testing our hypotheses about the neurocognitive mechanisms underlying belief updating. Future research should build on these findings by including participants from across the ideological spectrum to assess potential asymmetries and to determine whether the observed mechanisms generalize beyond left-wing belief systems.

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Second, our design focused on fMRI correlates and did not incorporate real-world behavioral outcomes (e.g., voting patterns, activism), which constrains conclusions about downstream consequences. Next, although our item-level analysis identified neural correlates of belief persistence, we did not examine individual-level predictors —such as socioeconomic status,

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conspiracy mentality, or need for cognitive closure—which may help explain within-group variability in openness to change. Investigating these factors will be an important avenue for future research. Although our study provides a context-sensitive test of whether resistance to belief change reflects a universal psychological mechanism or one that varies systematically across political cultures and information environments, it is important to note that Poland remains, broadly speaking, part of the Western cultural and political context, sharing many institutional and informational features with other Western democracies. Accordingly, the present findings should be interpreted as a step in the right direction toward broader generalization, but only a modest one, underscoring the need for future research conducted in non-Western and structurally distinct political settings.

Finally, we note that the sample in our study was not gender-balanced, with a predominance of female participants. Although the study was not designed to examine gender differences and our primary analyses did not focus on gender, this imbalance may limit the generalizability of the findings. Research on gender differences in belief updating and related cognitive processes shows mixed results, suggesting that while small differences exist in some cognitive biases, the overall magnitude is limited and context-dependent [27]. Therefore, although our sample is female-dominated, existing literature indicates that gender is unlikely to fundamentally alter the observed patterns of belief resistance. Nevertheless, future research should aim for more balanced recruitment to assess whether the observed effects replicate across genders and to explore potential gender-related differences in belief updating.

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Conclusion

Our study supports the idea that political beliefs are uniquely resilient because they are deeply integrated into the self, and this integration is reflected in distinct patterns of neural processing.

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The Default Mode Network appears to play a central role in this defensive response, whereas regions associated with cognitive flexibility and belief updating—such as the orbitofrontal cortex and dorsolateral prefrontal cortex—are more active when beliefs are less identity-laden. Together, these findings shed light on the neural and psychological architecture of belief change, offering a more nuanced understanding of why evidence so often fails to persuade.

Methods

Participants

The sample size was determined based on the study by Kaplan et al. [1] which suggests that for within-subject designs with a medium effect size ($d \approx 0.5$), approximately 20–30 participants are sufficient to achieve about 80% power in simple ROI contrasts. Forty-six participants (mean age = 25.39, SD = 7.86; range: 19–62 years; 9 male, 36 female, 1 nonbinary) were recruited via a social media platform to take part in the study. All participants self-identified as politically engaged, politically left-wing, and reported holding strong political and non-political beliefs. Specifically, as in original study, participants completed a screening questionnaire in which they were asked, 'Do you consider yourself a political person?' Responses ranged on a scale from 1 (not at all) to 5 (very much). They were also asked about their political identification with the question, 'Which of the following describes your political self-identification?' Responses ranged from 1 (strongly liberal) to 7 (strongly conservative). Additionally, participants rated their agreement with several political and non-political statements on a scale from 1 (strongly disagree) to 5 (strongly agree). We also verified whether participants were right-handed and had no history of psychological or neurological disorders. Inclusion criteria required selecting at least a 4 on the political engagement question, choosing 1, 2, or 3 on the 7-point political self-identification scale, and expressing agreement ('strongly agree,' 'agree,' or 'rather agree') with at least 8 political and 7 non-political

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statements. Participants also had to self-report being right-handed and having no history of psychological or neurological disorders.

Due to a technical error, behavioral data were unavailable for three participants, resulting in a final behavioral sample of 43 participants (mean age = 25.44 years, SD = 8.12; range = 18–62 years; 8 male, 34 female, 1 nonbinary). Approximately one week after the lab visit, participants received a follow-up questionnaire assessing their responses to the presented statements. Thirty-two participants completed this follow-up (mean age = 24.00 years, SD = 4.16; range = 19–37 years; 4 male, 27 female, 1 nonbinary).

Each participant received 100 PLN (approximately $30) per hour for their participation. All participants provided informed consent, and were informed of their right to withdraw from the study at any time. This study was conducted in accordance with the ethical principles outlined in the 1964 Declaration of Helsinki and was approved by the Institutional Review Board (#221.0042.61_2023). The preregistration for this study is available at: https://doi.org/10.17605/OSF.IO/9AKWX. All materials, scripts, and data necessary for replication are available on the OSF project page (for review https://osf.io/8gnz6/ view_only=24640477359546caa9f69137543f4e52).

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Stimuli

The experimental procedure closely followed that of Kaplan et al.[1], with the sole modification being the adaptation of certain stimuli to better reflect the Polish cultural and political context; for example, the issue of gun control—salient in the U.S. but not in Poland—was replaced with more locally relevant topics. This adapted procedure had been thoroughly tested and successfully implemented in a previous study [28].

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Each participant was presented with 8 political and 8 non-political statements that they had previously expressed strong agreement with. Each statement was followed by five challenges—brief counterarguments or pieces of evidence designed to oppose the original belief. The political statements addressed issues where participants were expected to hold identity-consistent views, such as "*Abortion should be legal*," "*Poland should agree to accept refugees*," and "*Sex education should be mandatory for every student.*" Full lists of the statements and the rationale for their inclusion are provided in the Supplementary Materials.

The non-political statements covered a diverse range of factual or commonly held beliefs, such as "*A college education generally improves a person's economic prospects*," "*Thomas Edison invented the light bulb*," and "*Reading ability in early childhood is a sign of intelligence.*"

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Experimental Procedure

Upon arrival for the fMRI session, participants received detailed instructions and were given the opportunity to ask the experimenter any questions. They then completed a practice trial designed to familiarize them with the task structure. This trial used a shortened version of the experimental sequence, featuring the statement "*Cats make better pets than dogs*" followed by three counterarguments.

Following the practice, each participant then completed four fMRI runs, each lasting 420 seconds, during which their beliefs were systematically challenged. At the start of each trial, a target statement appeared on the screen for 10 seconds, followed by a variable interstimulus interval of 4 to 6 seconds. Participants were instructed to press a response button once they had finished reading and comprehending the statement. They then viewed five counterarguments, each presented for 10 seconds, again pressing the button to indicate they had finished reading.

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After all five challenges, the original statement reappeared on the screen. Participants were given 12 seconds to rate how strongly they believed the statement, using an MRI-compatible response box in their right hand. They moved a cursor along a Likert scale ranging from 1 (*strongly disbelieve*) to 7 (*strongly believe*), which initially appeared at the midpoint. Each of the four runs included two political and two non-political statements. The order of statement types and the specific statements presented were randomized across participants. An overview of the procedure is illustrated in Figure 1.

Following the scanning session, participants completed a brief questionnaire assessing the overall credibility of the challenges and the extent to which they perceived them as threatening to their existing beliefs. These assessments captured general impressions of the stimulus set rather than item-by-item evaluations. Participants were then debriefed, informed that not all of the information presented during the study (in the form of challenges) was accurate, and thanked for their participation. Approximately one week later, they received a follow-up questionnaire via email, in which they rated their agreement with the same statements they had encountered during the scanning session.

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MRI Scanning

MRI data were acquired using a 3T Magnetom Prisma scanner (Siemens) equipped with a 64-channel head coil. High-resolution, whole-brain anatomical images were acquired using a T1-weighted MPRAGE sequence (192 sagittal slices; voxel size $0.9 \times 0.9 \times 0.9$ mm^3; TR = 2300 ms; TE = 2.32 ms; flip angle = 8°).

Functional images were obtained using a whole-brain echo-planar (EPI) pulse sequence developed by the Center for Magnetic Resonance Research (CMRR), University of Minnesota with the following parameters: 44 axial slices; $3 \times 3 \times 3$ mm isotropic voxels; TR = 1200 ms;

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TE = 27 ms; flip angle = 75°; multiband (MB) acceleration factor = 2; in-plane GRAPPA acceleration factor = 2; and phase encoding direction A >> P [29,30,31]. Four functional runs were acquired, each lasting approximately 7 minutes.

Details of MRI data preprocessing are provided in Section 2 of the Supplementary Materials.

fMRI Data Analysis

Preprocessed functional data were subsequently analyzed with AFNI software [32]. The analysis began with scaling to percent signal change and spatial smoothing (full-width at half-maximum = 6mm). Task activations were then estimated using generalized least squares time series fit with Restricted Maximum Likelihood (REML) estimation of the temporal auto-correlation structure (3dREMLfit toolbox). Each task component—statement, challenge, and rating—was modeled separately and convolved with a double-gamma hemodynamic response function (phase = 0 s). Time periods were defined as follows: statements, from onset to the offset of the statement; challenges, onset of the first to the offset of the fifth challenge for a given statement; ratings, onset of the second presentation of the statement to its offset after the rating. Political and non-political trials were modeled separately, resulting in six-task component regressors in total. Six motion-correction parameters were also included as nuisance regressors. A statistical map was generated for each subject, and these subject-level maps were entered into a group-level analysis using 3dttest++.

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Following the approach of Kaplan and colleagues, we examined the relationship between brain activity elicited by reading the challenges and the degree of belief change using two complementary analyses: (i) a whole brain item-wise analysis and (ii) a subject-wise analysis of *a priori* selected regions of interest (ROIs) in the amygdala and insular cortex. For the whole-brain analysis, each run was first modeled with eight regressors, each corresponding to a specific statement or a specific challenge period (note that each run included the presentation of four

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statements, each followed by a series of challenges). Task component periods were specified as in the previous analysis and convolved with a double-gamma hemodynamic response function (phase = 0 s). Six motion parameters were also included as nuisance regressors. We then computed brain-activity maps for each specific stimulus item, combining responses to specific individual statements or challenges across all participants. These item-level activity maps were tested for correlations with item-wise belief change, with scores averaged across participants for each item separately.

For the subject-wise ROI analysis, we focused on the amygdala and insular cortex due to their well-established roles in emotion and affective processing. ROIs were defined following the procedure used by Kaplan and colleagues. The amygdala was defined using the Harvard-Oxford subcortical probability atlas. Masks for the insular ROIs—comprising the dorsal anterior, ventral anterior, and posterior insula—were defined according to the cluster analysis performed by Deen and colleagues [33]. For each ROI, beta values corresponding to the challenge periods were extracted from the GLM analysis. These beta values were then tested for correlations with the subject-wise belief change scores (averaged across items for each participant separately). Owing to the fact that changes in political belief ratings were minimal, only scores for non-political statements were included in this analysis. Beta values and subject-wise belief change were first tested for normality with a Shapiro-Wilk test, and depending on the results, correlations were computed using either Pearson's correlation coefficient (for normally distributed data) or Spearman's rank correlation coefficient (for non-normal distributions). In order to assess the plausibility of both null and experimental hypotheses, classical statistical tests were supplemented with Bayesian inference conducted in JASP (version 0.19.3; [34]). Results are reported as Bayes factors for the null hypothesis (BF_{01}), defined as the ratio of the probability of the observed data

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under the null hypothesis (H_0) to that under the alternative hypothesis (H_1). Interpretation of BF_{01} values provided in the Results section follows the guidelines[35].

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Author contributions

MK: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. GC: Writing – review & editing; PS: Writing – review & editing, Writing – original draft (Co-drafted the results section), Methodology, Behavioral Formal Analyses, Data curation. SH: Writing – review & editing, Methodology, Investigation; ADP: Writing – review & editing, Writing – original draft (Co-drafted the fMRI results section), fMRI Formal Analyses, fMRI Data curation. ŁD: Writing – review & editing; Writing – original draft (Co-drafted the fMRI results section), fMRI Formal Analyses; fMRI Data curation.

Data availability statement (mandatory)

The preregistration for this study is available at: https://doi.org/10.17605/OSF.IO/9AKWX. All materials, scripts, and data necessary for replication are available on the OSF project page (for review https://osf.io/8gnz6/?view_only=24640477359546caa9f69137543f4e52).

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Additional Information (including a Competing Interests Statement)

The authors declare no conflicts of interest.

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Figure legends (these are limited to 350 words per figure)

Figure 1. Across four fMRI scans, participants completed four trials per scan, each comprising two politically themed and two non-political stimulus sets. In each trial, participants were first shown a statement representing a belief they had previously endorsed, followed by five counterarguments challenging that belief. At the end of each trial, the original statement reappeared, and participants rated the strength of their belief on a scale from 1 (*strongly disbelieve*) to 7 (*strongly believe*).

Figure 2. Change in belief strength for political and non-political statements before and after the scanning session.

Figure 3. Brain activation during challenges to political versus non-political beliefs. Red/yellow regions indicate greater activation when processing challenges to political beliefs (P > NP). Blue/green regions indicate greater activation when processing challenges to non-political beliefs (NP > P).

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Figure 4. Relationship between belief change and brain activity. (A) Left temporal pole; (B) dorsomedial prefrontal cortex; (C) inferior temporal gyrus. Scatter plots depict the relationships observed in the peak voxels of each region. (D) Stimulus items ranked by average belief-change score.

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Tables

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Table 1. Peak activations for Political vs. Non-political (P>NP) and Non-political vs. Political (NP>P) brain activations during challenges .

Region	Side	Z-value	x	y	z
NP>P			------	------	------
Orbitofrontal Cortex	L	3.33	-42.0	+51.0	-18.0
Dorsolateral Prefrontal Cortex	P	3.78	+54.0	+18.0	+45.0
Orbitofrontal Cortex	P	3.75	+42.0	+63.0	-9.0
Inferior Temporal Lobe	L	3.22	-66.0	-54.0	-6.0
Presupplementary motor area	L/R	5.25	+3.0	+33.0	+45.0
Superior Parietal Lobe	R	3.79	+36.0	-69.0	+57.0
Superior Parietal Lobe	L	3.71	-30.0	-75.0	+57.0
Inferior Temporal Lobe	R	3.89	+69.0	-36.0	-9.0
Temporal Pole	L	3.51	-21.0	+3.0	-42.0
P>NP.					
Medial Prefrontal Cortex	L/R	4.25	0.0	+54.0	-3.0
Inferior Parietal Lobe / Supramarginal Gyrus	R	3.13	-69.0	+24.0	+21.0
Posterior Cingulate Cortex / Precuneus	L/R	3.85	+3.0	+51.0	+27.0

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Figures

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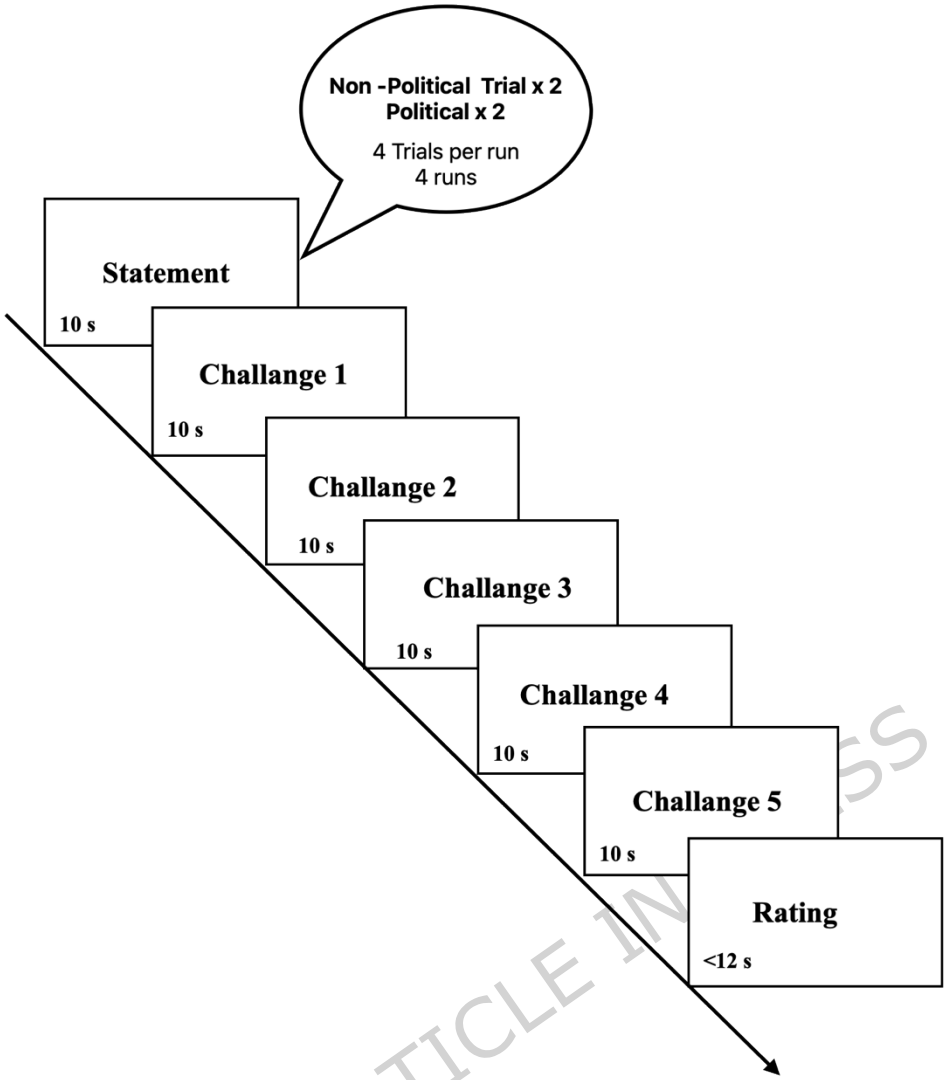


Figure 1. Across four fMRI scans, participants completed four trials per scan, each consisting of two politically themed and two non-political stimulus sets. In each trial, participants were first presented with a statement representing a belief they had previously endorsed. This was followed by five counterarguments challenging that belief. At the end of each trial, the original statement reappeared, and participants rated the strength of their belief on a scale from 1 (strongly disbelieve) to 7 (strongly believe).

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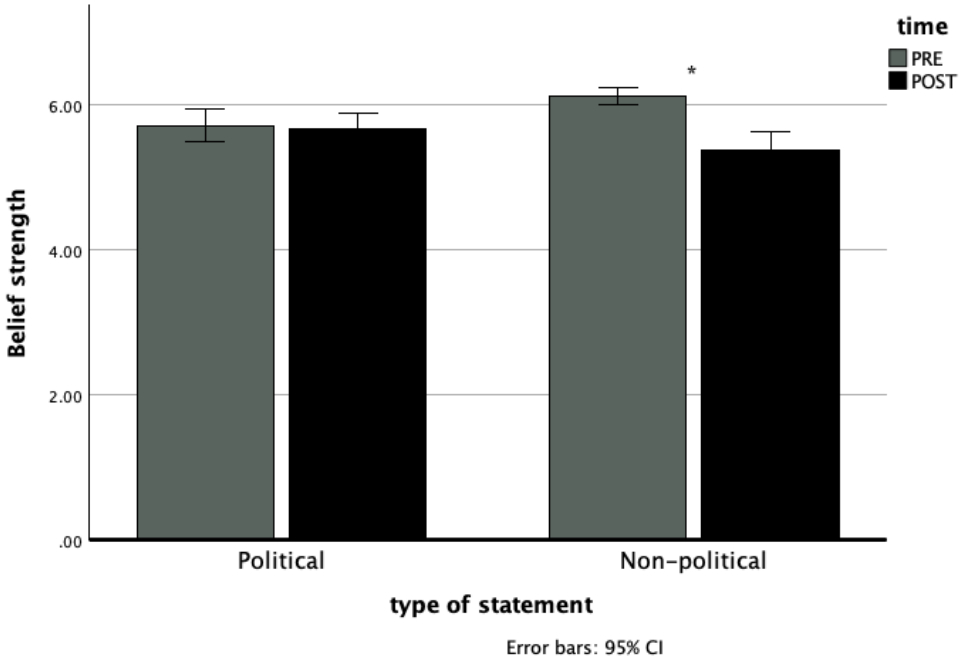


Figure 2. Change in belief strength for both political and non-political statements before and after the scanning session.

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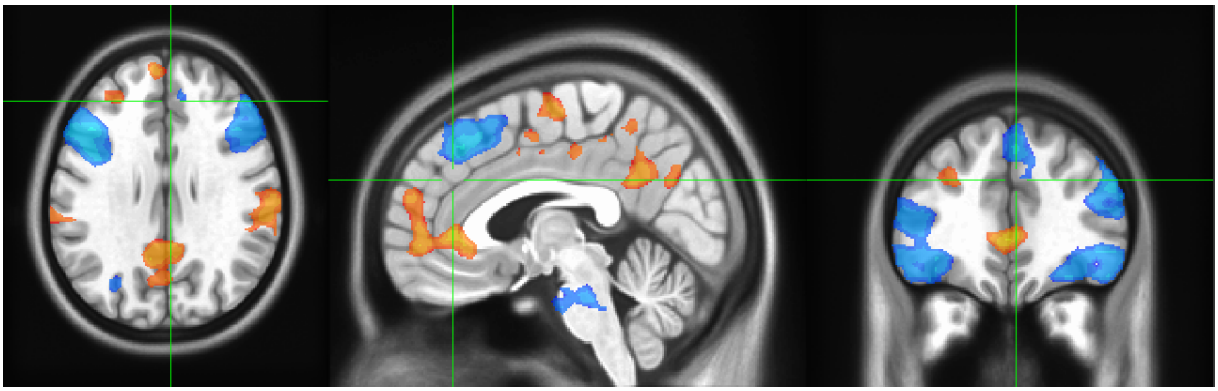


Figure 3. Brain activation during challenges to political vs. non-political beliefs. In red/yellow, brain regions that showed increased signal while processing challenges to political beliefs (P > NP). In blue/green, brain regions that showed increased signal during challenges to non-political beliefs (NP > P).

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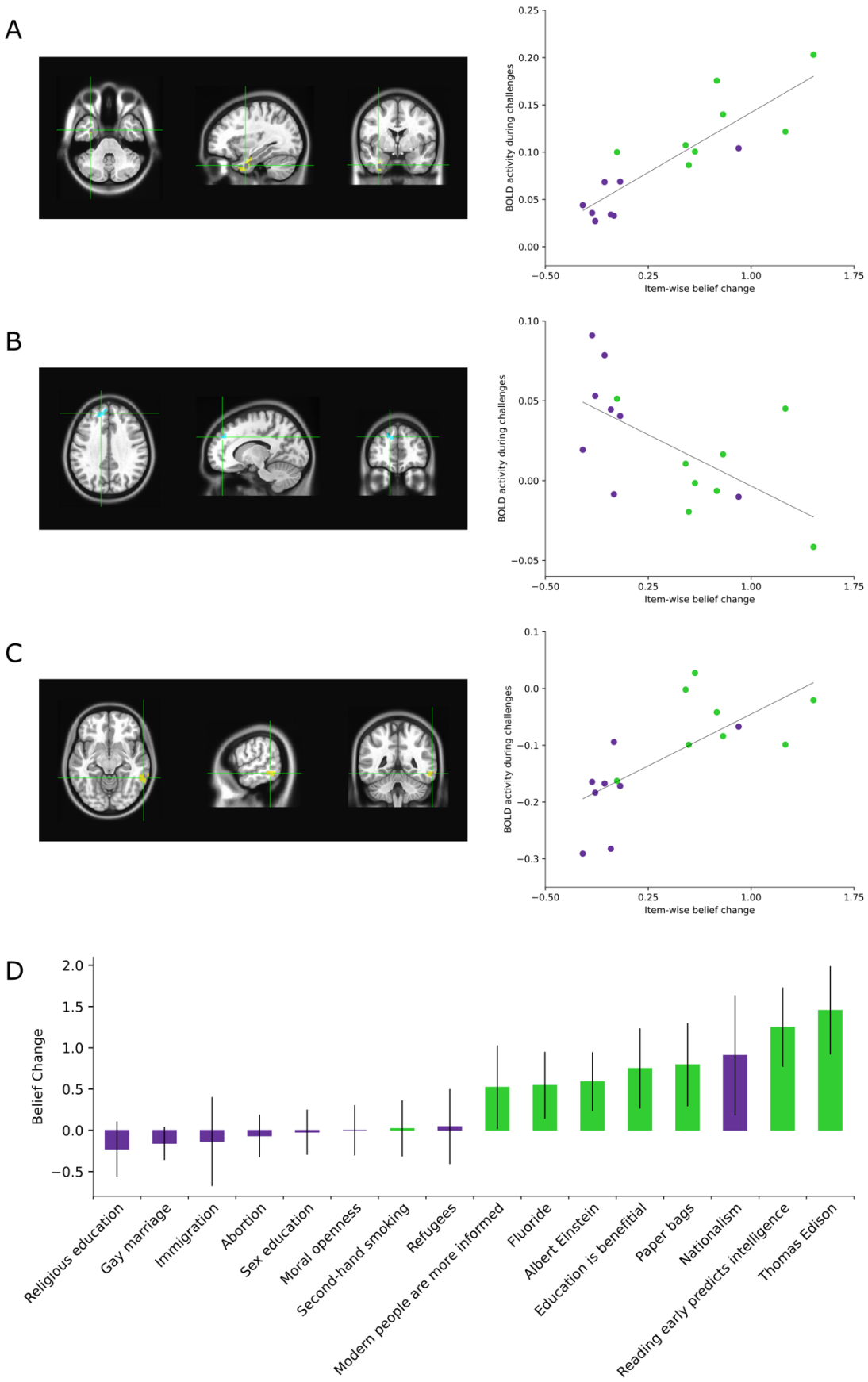


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